

Tackling Non-Revenue Water for a Water-Secure Viksit Bharat



For Prelims: Urban Local Bodies , Geographic Information Systems , Wetlands, Urban flooding , AMRUT 2.0 , Smart Cities Mission

For Mains: Water resource management and urban water security, Circular economy and wastewater reuse.

Source: TH

Why in News?

India's ambition to build a 'Viksit Bharat' relies heavily on its urban centres , which are increasingly facing acute water stress due to **groundwater over-exploitation , rapid urbanization, and climate change** , bringing urban water governance into sharp policy focus.

- A major policy concern is that **urban utilities lose more than 30% of the water they supply as Non-Revenue Water (NRW)** , undermining municipal finances, reducing service efficiency, and threatening long-term urban water security.

Summary

- India's urban water crisis is driven by high Non-Revenue Water (NRW), groundwater depletion, rapid urbanisation, climate change, and fragmented governance, threatening water security and the vision of Viksit Bharat.
- Achieving sustainable urban water governance requires reducing NRW, promoting wastewater reuse, strengthening Urban Local Bodies (ULBs), adopting smart water technologies, conserving groundwater, and implementing integrated, climate-resilient water management aligned with SDG 6.

What is Non-Revenue Water (NRW)?

- **Non-Revenue Water (NRW):** It refers to water that has been treated and pumped into the municipal distribution system but is **"lost" before it reaches the consumer or is billed**. It primarily consists of:
 - **Physical (Real) Losses:** Water lost due to leakages from aging pipelines, broken valves, and overflowing reservoirs.
 - **Commercial (Apparent) Losses:** Water lost due to inaccurate metering, unauthorized connections (theft), and unbilled public consumption (such as free public standposts).
 - In advanced global cities, **NRW is maintained strictly under 10%**. Conversely, in most Indian metropolitan areas, it ranges between **40% to 50%** .
 - This means nearly half the water treated at a **high cost yields zero municipal revenue and never reaches the tap**.

Challenges in Urban Water Governance

- **Financial Unsustainability:** Due to high NRW, **Urban Local Bodies (ULBs)** currently recover less than **55% of their operation and maintenance (O&M) costs**.
 - This lack of cost recovery cripples their ability to repair aging infrastructure and expand services.
- **Over-reliance on Distant Sources:** Cities increasingly depend on **distant rivers and deep aquifers** , leading to high energy costs for pumping, inter-state water disputes, and severe vulnerability to droughts.
- **Infrastructural Deficit & Linear Use:** Only about **28% of urban wastewater is treated** .
 - Most cities operate on a **"use and discard" linear model** rather than a **circular economy** , leaving massive potential for wastewater reuse untapped.
 - The absence of **Geographic Information Systems (GIS) -based pipeline mapping, SCADA (Supervisory Control and Data Acquisition) systems, and District Metered Areas (DMAs)** prevents proactive water management. Utilities largely respond only after leaks or pipeline failures occur, resulting in persistent water losses and operational inefficiencies.
- **Inequitable Distribution:** The current system disproportionately harms the urban poor. Those in informal settlements often pay a premium for **private tankers or lose income due** to time spent collecting water, while connected, affluent households enjoy subsidized, flat-rate piped water.
- **Institutional Fragmentation:** Urban water infrastructure is typically developed by **state-level parastatal agencies** but operated by **ULBs** .
 - This fragmented governance creates an accountability gap, while financially and technically weak ULBs struggle to maintain complex infrastructure.
- **Ecologically Unsustainable Urban Planning:** Encroachment of **wetlands, lakes, floodplains, and groundwater recharge zones** has weakened cities' natural water systems. This ecological degradation simultaneously increases **urban flooding during monsoons** and **water scarcity during summers** , reducing long-term urban water resilience.

Major Government Initiatives for Urban Water Governance

- **AMRUT 2.0 (Atal Mission for Rejuvenation and Urban Transformation)**
- **Smart Cities Mission (SCM)**
- **Jal Jeevan Mission (Urban)**
- **Pey Jal Survekshan**
- **Composite Water Management Index (CWMI)**
- **Jal Shakti Abhiyan: Catch the Rain**

What Measures are Needed for Effective Urban Water Governance?

- **Mandating 'Sponge City' Infrastructure:** Move beyond superficial rainwater harvesting by embedding **green-blue infrastructure** (permeable pavements, restored peri-urban wetlands, and bioswales) into the **city's master plan**. This decentralizes aquifer recharge and dual-functions as flood mitigation.
- **Statutory Decoupling of Water Use:** Legally mandate the separation of potable and non-potable supply chains.
 - Enforce **"Zero Liquid Discharge" (ZLD) protocols** for heavy industries and large commercial complexes, forcing them to rely exclusively on recycled municipal effluent for cooling and construction.
- **Pricing Groundwater Extraction:** Introduce strict volumetric pricing on **private borewells in affluent neighborhoods and commercial zones** to disincentivize the rapid depletion of local hydro-geological buffers.
- **GIS and SCADA Integration:** Shift from reactive fire-fighting (fixing visible leaks) to predictive network management.
 - Mandate the digitization of underground assets using **Geographic Information Systems (GIS) paired with SCADA (Supervisory Control and Data Acquisition)** to monitor real-time pressure anomalies.
- **Ring-Fencing NRW Budgets via DMAs:** Break down monolithic city networks into hydraulically isolated **District Metered Areas (DMAs)**.
 - State finance commissions must strictly tie **devolution of funds to verifiable reductions in Non-Revenue Water (NRW)** in these high-loss zones.
- **Concurrent Sewer Auditing:** Operational efficiency must include waste. Launch synchronized audits of **sewer networks to arrest sewage exfiltration**, which is the primary cause of **groundwater contamination and subsequent public health crises (e.g., urban cholera/typhoid outbreaks)**.
- **Formalizing the Informal Water Market:** Mandate municipal licensing, GPS tracking, and quality testing for tankers, converting them into a regulated emergency utility arm rather than a predatory parallel economy.
- **Redefining Service Benchmarks:** Shift the metric of success from "kilometers of pipeline laid" to "equitable hydrological security."
 - Prioritize capital expenditure on localized pressure management and decentralized **smart-standposts in informal settlements to eliminate the "time-poverty"** tax disproportionately borne by women.

- **Institutional Convergence via Water Regulatory Authorities:** Establishing **statutory State Urban Water Regulatory Authorities** (akin to State Electricity Regulatory Commissions) to mandate NRW reduction targets, standardize energy audits of pumping stations, and depoliticize tariff revisions.
- **Leveraging Hybrid Annuity Models (HAM):** Utilizing Public-Private Partnerships under HAM to incentivize private players to plug distribution losses.
 - Annuity payments to the concessionaire should be strictly linked to verifiable reductions in the baseline NRW over a long-term concession period.

Conclusion

India cannot build a **Viksit Bharat on a leaky water system**. Moving from crisis management to **water resilience** requires integrated urban planning, efficient water governance, wastewater reuse, and equitable access. Strengthening urban water security is essential for climate-resilient growth and achieving **SDG 6 (Clean Water and Sanitation)**, while advancing the broader vision of sustainable and inclusive urban development.

Drishti Mains Question:

What is Non-Revenue Water (NRW)? Discuss its implications for urban governance and suggest measures to reduce NRW in Indian cities.

Frequently Asked Questions (FAQs)

1. What is Non-Revenue Water (NRW)?

Non-Revenue Water (NRW) refers to **treated water that is lost before reaching consumers or being billed** due to pipeline leakages, unauthorized connections, faulty metering, and unbilled public consumption.

2. What are District Metered Areas (DMAs)?

District Metered Areas (DMAs) are hydraulically isolated zones within a water distribution network that help **detect leaks, manage pressure, and reduce Non-Revenue Water (NRW)**.

3. What is Zero Liquid Discharge (ZLD)?

Zero Liquid Discharge (ZLD) is a wastewater management approach that requires industries to **treat, recycle, and reuse wastewater**, minimizing freshwater use and preventing liquid effluent discharge.

4. What is SCADA in urban water management?

Supervisory Control and Data Acquisition (SCADA) is a real-time monitoring system used to track water flow, pressure, and leakages, enabling efficient operation of urban water supply networks.

5. What is a Sponge City?

A **Sponge City** is an urban planning approach that uses **green-blue infrastructure** such as wetlands, bioswales, and permeable pavements to absorb rainwater, recharge groundwater, and reduce urban flooding.

[Watch Video on YouTube:

▶ https://www.youtube.com/embed/N6Ls0_t8jYE]

UPSC Civil Services Examination, Previous Years Questions(PYQs)

Prelims:

Q.1. Which one of the following ancient towns is well known for its elaborate system of water harvesting and management by building a series of dams and channelizing water into connected reservoirs? (2021)

- (a) Dholavira
- (b) Kalibangan
- (c) Rakhigarhi
- (d) Ropar

Ans: A

Q2. With reference to 'Water Credit', consider the following statements: (2021)

1. It puts microfinance tools to work in the water and sanitation sector.
2. It is a global initiative launched under the aegis of the World Health Organization and the World Bank.
3. It aims to enable the poor people to meet their water needs without depending on subsidies.

Which of the statements given above are correct?

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

Ans: C

Mains :

Q. What are the salient features of the Jal Shakti Abhiyan launched by the Government of India for water conservation and water security? **(2020)**

Q. Suggest measures to improve water storage and irrigation system to make its judicious use under the depleting scenario. **(2020)**

Q. Account for the huge flooding of million cities in India including the smart ones like Hyderabad and Pune. Suggest lasting remedial measures. **(2020)**



Source: TH

Pickaxe Mountain in Iran has gained global attention after US threatened military strikes, amid reports that the mountain may conceal deep underground nuclear facilities linked to Iran's nuclear programme.

- The developments have also revived discussions on **India's decision not to launch a pre-emptive strike on Pakistan's Kahuta nuclear facility** in the 1980s despite strategic considerations.
- **Geographical Setting:** Situated roughly **220 km south of Tehran** near **Natanz** , Pickaxe Mountain is a prominent ridge within the **Karkas Range (Reshteh Kuh-e Karkas)** , which spans over 100 km between **Kashan and Ardestan** in an arid plateau.
- **Tectonic Origin:** The range forms part of the **Urumieh-Dokhtar magmatic belt** , created through the **subduction of the Neo-Tethyan Ocean** under the Iranian continental plate prior to the **Arabia-Eurasia continental collision** over the last 66 million years.
- **Rock Composition (Lithology):** The bedrock is predominantly made of mechanically strong volcanic rocks (including **andesite, rhyolite, and dacite**) interbedded with **pyroclastic deposits and volcanic tuffs** .
- **Structural Complexity & Wave Dispersion:** Extensive network of **fault zones, fractures, and varied rock densities** scatter seismic and kinetic shock waves, making military penetration and blast modelling unpredictable.
- **Strategic Defense & Hardening:** The high energy required to fracture intact volcanic rock combined with deep-seated natural subterranean shielding complicates kinetic strikes, offering Iran an **asymmetrical strategic advantage** alongside control near key maritime chokepoints like the **Strait of Hormuz** .



India's Kahuta Dilemma and Nuclear Restraint

- **Pokhran-I (1974)** (**Smiling Buddha**) marked **India's first nuclear test** , prompting Pakistan to

accelerate its nuclear programme centred around the **Kahuta uranium enrichment facility** .

- In **1981** , the **Indian Air Force studied the feasibility of a pre-emptive strike on Kahuta** , though no official plan was confirmed.
- India ultimately chose **strategic restraint** over military action, allowing Pakistan to achieve **nuclear capability by the late 1980s** , which reshaped South Asia's strategic balance.
- To reduce the risk of attacks on nuclear facilities, **India and Pakistan signed the Agreement on the Prohibition of Attack against Nuclear Installations and Facilities in 1988** .
 - Since **1992** , both countries have **annually exchanged lists of their nuclear installations** .
- Today, India follows a policy of **credible minimum deterrence** and **strategic restraint** , while strengthening its **multi-layered air and missile defence** through projects such as **Project Kusha** and **Sudarshan Chakra** .

Read more: [Conflict Zones in the US - Israel - Iran War](#)

Bank of Baroda Data Breach on Dark Web



Source: TH

Public sector lender Bank of Baroda (BoB) has initiated a forensic investigation following an alleged data breach that reportedly leaked around 1 TB of **customer-related data** onto the **dark web** , raising fresh concerns over cybersecurity in the banking sector.

- The bank stated that **its core banking systems were not compromised** and continue to remain secure.
- Cyber incidents, such as the **Tata Electronics ransomware attack (2026)**, **BSNL data breach (2024)**, **WazirX crypto heist (2024)**, **ICMR data leak (Dark Web, 2023)**, **Air India cyberattack (2021)**, and the latest Bank of Baroda data breach, have raised serious concerns over India's cybersecurity preparedness, data privacy, critical information infrastructure protection, and cyber resilience.
- **Dark Web:** The World Wide Web consists of three broad layers:
 - **Surface Web:** Publicly accessible and indexed by search engines (e.g., normal websites, news sites).
 - **Deep Web:** Unindexed but legitimate parts of the **web requiring authentication** (e.g., banking portals, private email inboxes, corporate databases).
 - **Dark Web:** A small, intentionally hidden subset of the Deep Web that requires **specific software (like the Tor browser)** to access.
 - It anonymizes user traffic and IP addresses, making it a safe haven for **illegal marketplaces, cybercrime, and data trafficking**.

India's Framework for Cyber Security and Data Protection

- **CERT-In (Indian Computer Emergency Response Team):** The nodal agency to **deal with cyber security threats like hacking and phishing** .
 - Financial institutions are mandated to report severe cyber incidents to CERT-In within 6 hours.
- **RBI's Cyber Security Framework for Banks:** Mandates banks to formulate a board-approved cyber-security policy, set up **a Cyber Crisis Management Plan (CCMP)**, and report unusual cyber incidents immediately.
- **Digital Personal Data Protection (DPDP) Act, 2023:** Imposes strict obligations on **Data Fiduciaries (like banks)** to protect personal data.
 - In case of a personal data breach, significant penalties can be levied for failing to deploy reasonable security safeguards.
- **IT Act, 2000:** Section 43A and Section 66 penalize unauthorized access and the failure to protect sensitive personal data.

Surface Web vs. Deep Web vs. Dark Web

Surface Web

- Searchable websites
- E-commerce sites
- News sites
- Public blogs and forums



Deep Web

- Websites requiring a login
- Email accounts
- Online banking accounts
- Subscription services



Dark Web

- Sites requiring access through Tor
- Websites not indexed on search engines
- .onion domains
- Decentralized marketplaces



BHAVYA Rasayan Scheme



Source: PIB

The Union Cabinet, chaired by the Prime Minister, has approved the **Bharat Audyogik Vikas Yojana Rasayan (BHAVYA Rasayan) Scheme** for establishing **three dedicated Chemical Parks** across India.

- **About:** Announced in the [Union Budget 2026-27](#) , **BHAVYA Rasayan** aims to establish three dedicated Chemical Parks with integrated infrastructure for chemical, petrochemical, downstream and ancillary industries.
- **Funding:** The scheme has a **total outlay of Rs. 3,030 crore**, comprising Rs. 3,000 crore for common infrastructure and basic utilities and Rs. 30 crore for administrative expenditure.
- **Centre-State Contribution:** The Central Government will provide up to **Rs. 1,000 crore per park**, subject to a minimum contribution of Rs. 500 crore from the concerned State Government.
- **Implementation:** The scheme will be implemented for five years, from **FY 2026-27 to FY 2030-31**, with the parks developed by State Governments through the Challenge Route.
- **Land Requirement:** Each Chemical Park must have **at least 8 sq. km. or 2,000 acres of contiguous and encumbrance-free land** .
- **Infrastructure:** The parks will provide **plug-and-play facilities** such as effluent **treatment plants**, [hazardous-waste management systems](#) , **water supply**, **solvent recovery units**, **steam networks**, **pipelines**, **logistics** and warehousing.
- **Economic Benefits:** The scheme will reduce **logistics and production costs**, **improve resource efficiency**, **attract domestic and foreign investment**, **increase exports**, **promote import substitution** and strengthen India's integration into global value chains.
- **Significance:** Sustainable infrastructure and better environmental compliance will support employment generation, boost downstream sectors and advance the goals of [Atmanirbhar Bharat](#) and [Viksit Bharat @ 2047](#) .

Read More: [Bharat Audyogik Vikas Yojna](#)